

Q. Convert SOP to POS expression.

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$
$$= 000 + 010 + 011 + 101 + 111$$

$$Y = m_0 + m_2 + m_3 + m_5 + m_7.$$

$$Y = \sum (0, 2, 3, 5, 7) \rightarrow \text{min terms.}$$

$$Y = \pi (1, 4, 6) \rightarrow \text{max terms.}$$

$$Y = M_1 + M_4 + M_6$$

$$Y = 001 + 100 + 110$$

$$Y = (A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$$

Convert POS to SOP expression:

$$Y = (A+B+\bar{C})(\bar{A}+B+C)(\bar{A}+\bar{B}+C)$$
$$= \underbrace{001}_1 \quad \underbrace{100}_4 \quad \underbrace{110}_6$$

$$Y = M_1 + M_4 + M_6$$

$$Y = \pi(1, 4, 6) \rightarrow \text{maxterms}$$

$$Y = \Sigma(0, 2, 3, 5, 7) \rightarrow \text{minterms}$$

$$Y = m_0 + m_2 + m_3 + m_5 + m_7$$

$$Y = 000 + 010 + 011 + 101 + 111$$

$$Y = \bar{A}\bar{B}\bar{C} + \bar{A}B\bar{C} + \bar{A}BC + A\bar{B}C + ABC$$

NOTE: For conversion between POS and SOP, in case of standard forms, first convert it from standard to canonical and then convert from either SOP to POS or POS to SOP as required.